ACTION POTENTIAL

Dr Shahneela Siraj

Associate Professor

DIKIOHS,DUHS.

LEARNING OBJECTIVES

- At the end of the lecture ,students should be able to
- Discuss post synaptic potentials (PSP’s)
- Define Action Potential
- Explain key properties of Action potentials
- Describe phases of Action Potentials
- Explain Refractory period.

BASIC ELECTROPHYSIOLOGICAL TERMS

- **Excitability**: The ability of the cell to generate the action potential

- **Excitable cells**: Cells that generate action potential during excitation.
  - in excitable cells (muscle, nerve, secretory cells), the action potential is the **marker** of excitation.
- **Stimulus**: a sudden change of the (internal or external) environmental condition of the cell.
  - includes physical and chemical stimulus.
  - The electrical stimulus is often used for the physiological research.

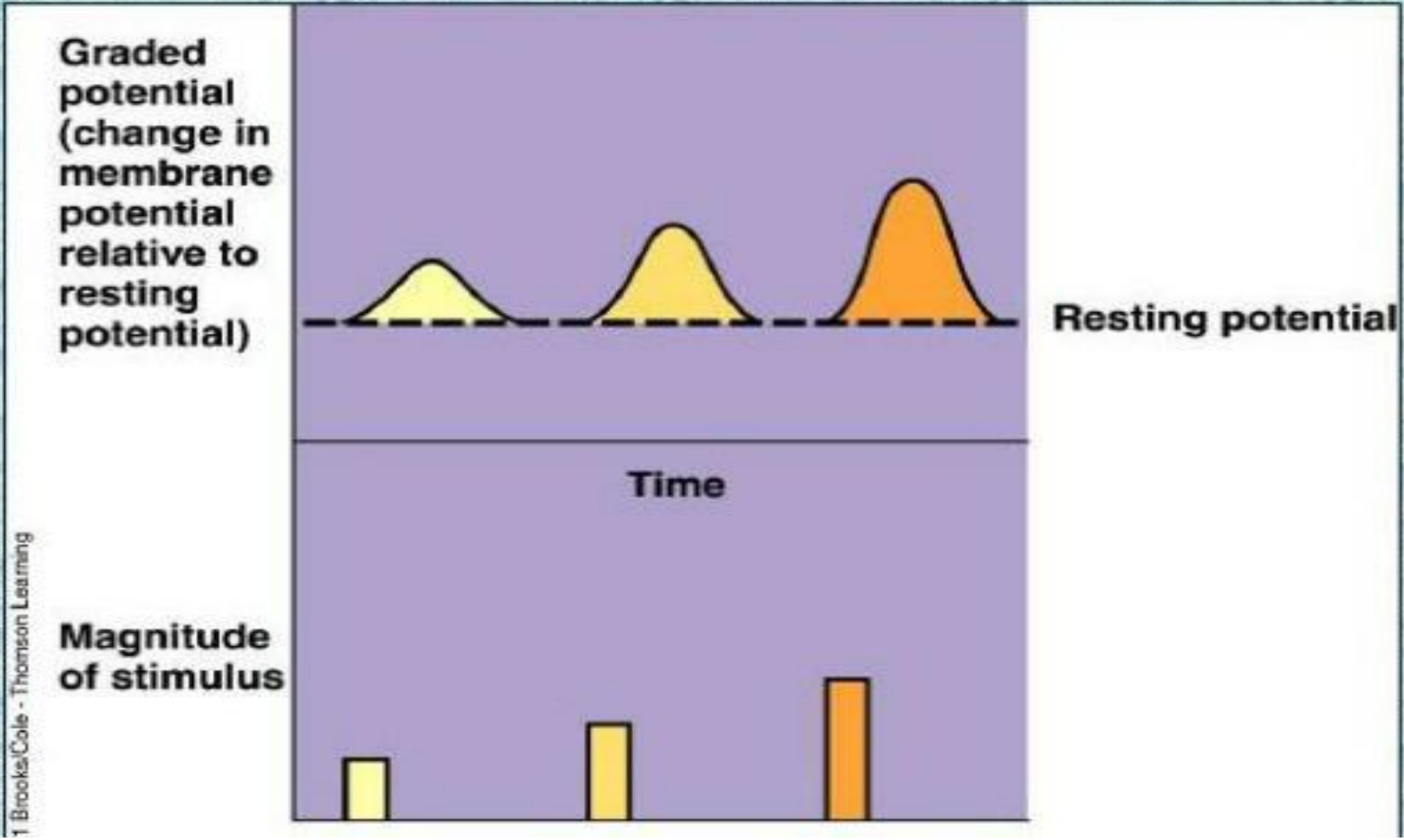
- **Threshold** (intensity): the lowest or minimal intensity of stimulus to elicit an action potential
  - (Three factors of the stimulation: intensity, duration, rate of intensity change)

CONT

- **Types of stimulus:**

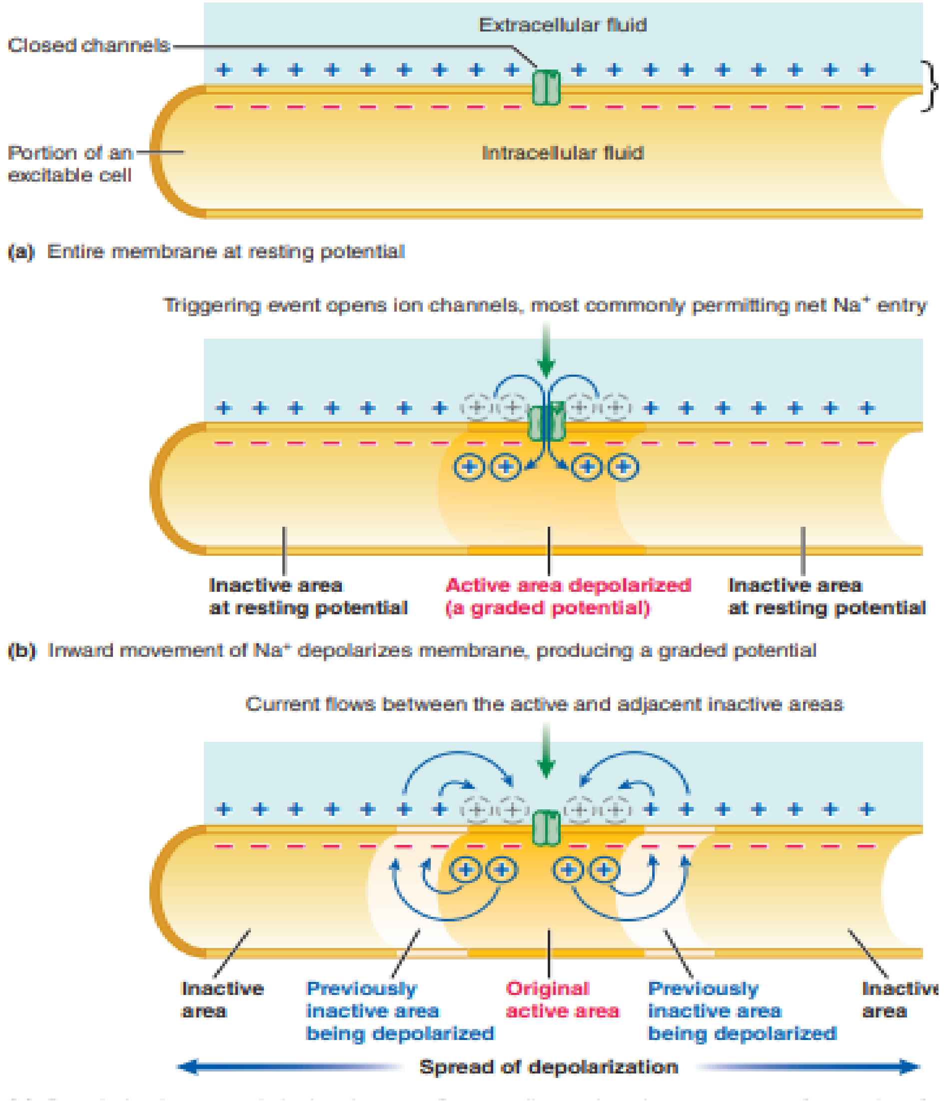
  - **Threshold stimulus**: The stimulus whose strength or voltage is sufficient to excite the tissue is called Threshold or minimal stimulus.
  - **Subthreshold stimulus**: The stimulus with the intensity weaker than the threshold.
  - **Suprathreshold stimulus**: The stimulus with the intensity greater than the threshold.

Graded Potentials



Graded Potentials/Electrotonic potential

- Graded potentials are local changes in membrane potential.
- For example, membrane potential could change from –70 to –60 mV (a 10-mV graded potential)
- Graded potentials are usually produced by a specific triggering event that causes gated ion channels to open in a specialized region of the excitable cell membrane.
- The resultant ion movement produces the graded potential, which most commonly is a depolarization resulting from net Na entry. The graded potential is confined to this small, specialized region of the total plasma membrane.
- The stronger the triggering event, the larger the resultant graded potential.
- The stronger the triggering event is, the more gated Na/Ca channels open,more positive charges in the form of Na enter the cell.
- The longer the duration of the triggering event, the longer the duration of the graded potential.
- The following are all graded potentials: post-synaptic potentials, receptor potentials, end-plate potentials, pacemaker potentials, and slow-wave potentials.



EXCITATORY POST SYNAPTIC POTENTIALS

- Excitatory postsynaptic potentials (EPSPs) are synaptic inputs that depolarize the postsynaptic cell, bringing the membrane potential closer to threshold and closer to firing an action potential.
- EPSPs are produced by opening Na+ and K+ channels, Opening of ion channels which leads to **depolarization** makes an action potential *more likely*, hence “excitatory PSPs”: **EPSPs.**
- **Inside becomes less negative**

  Caused by

  - **Na^+ channels (influx)** , **Ca^{2+} channels(influx)**
  - Excitatory neurotransmitters include ACh, norepinephrine, epinephrine, dopamine, glutamate & serotonin.

INHIBITORY POST SYNAPTIC POTENTIALS

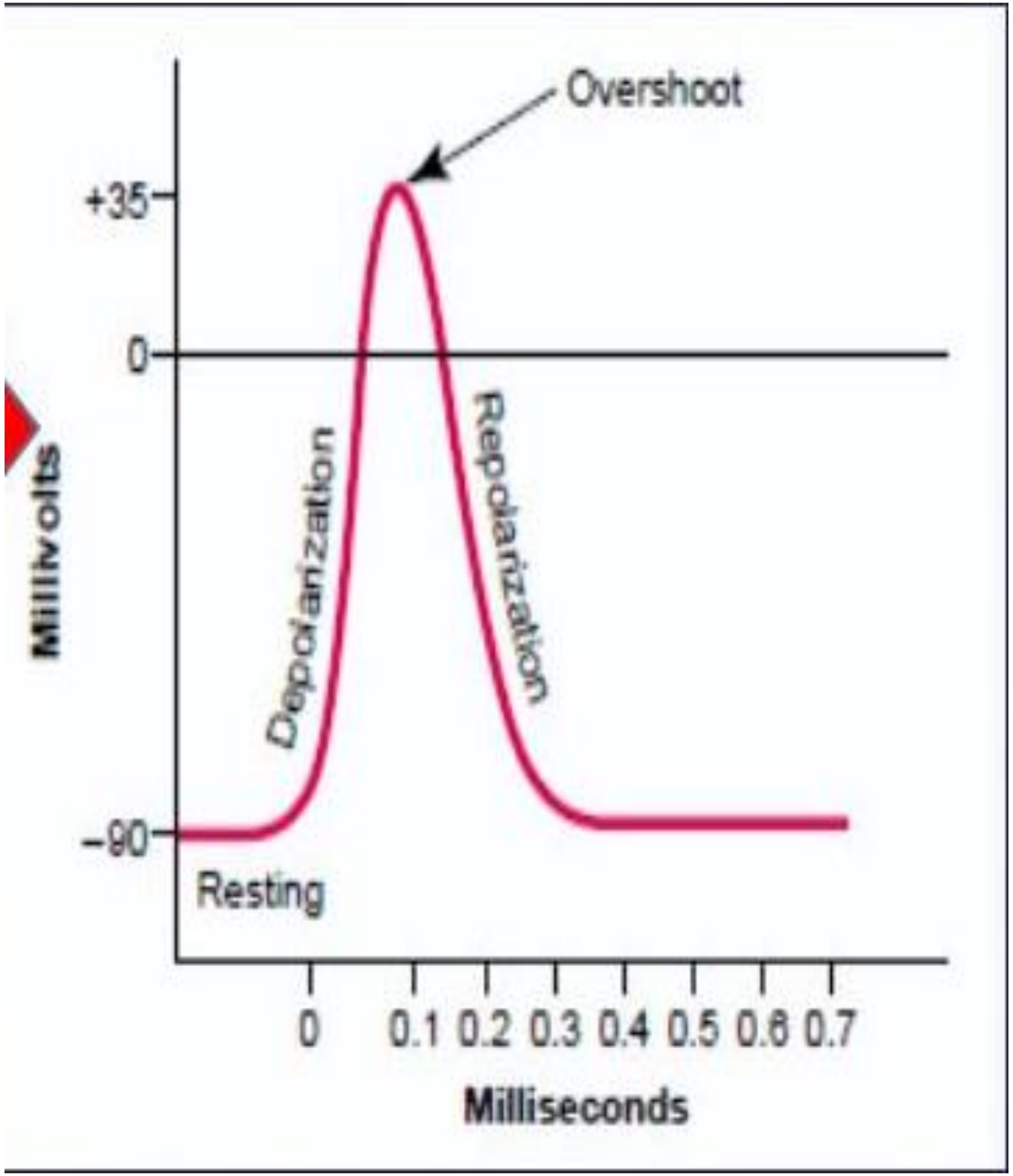
- Inhibitory postsynaptic potentials (IPSPs) are synaptic inputs that hyperpolarize the postsynaptic cell, taking the membrane potential away from threshold and farther from firing an action potential.
- IPSPs are produced by opening Cl– channels, Opening of ion channels which leads to **hyperpolarization** makes an action potential *less likely*, hence "inhibitory PSPs": **IPSPs.**
  - Inside of post-synaptic cell becomes **more negative**.
  - Caused by opening of
  - **K^+ channel (outflux), Cl channel(influx)**
  - Inhibitory neurotransmitters are g-aminobutyric acid (GABA) and glycine.

TABLE 31.1: Properties of action potential and graded potential

Action Potential	Graded potential
Propagative	Non-propagative
Long-distance signal	Short-distance signal
Both depolarization and repolarization	Only depolarization or hyperpolarization
Obeys all-or-none law	Does not obey all-or-none law
Summation is not possible	Summation is possible
Has refractory period	No refractory period

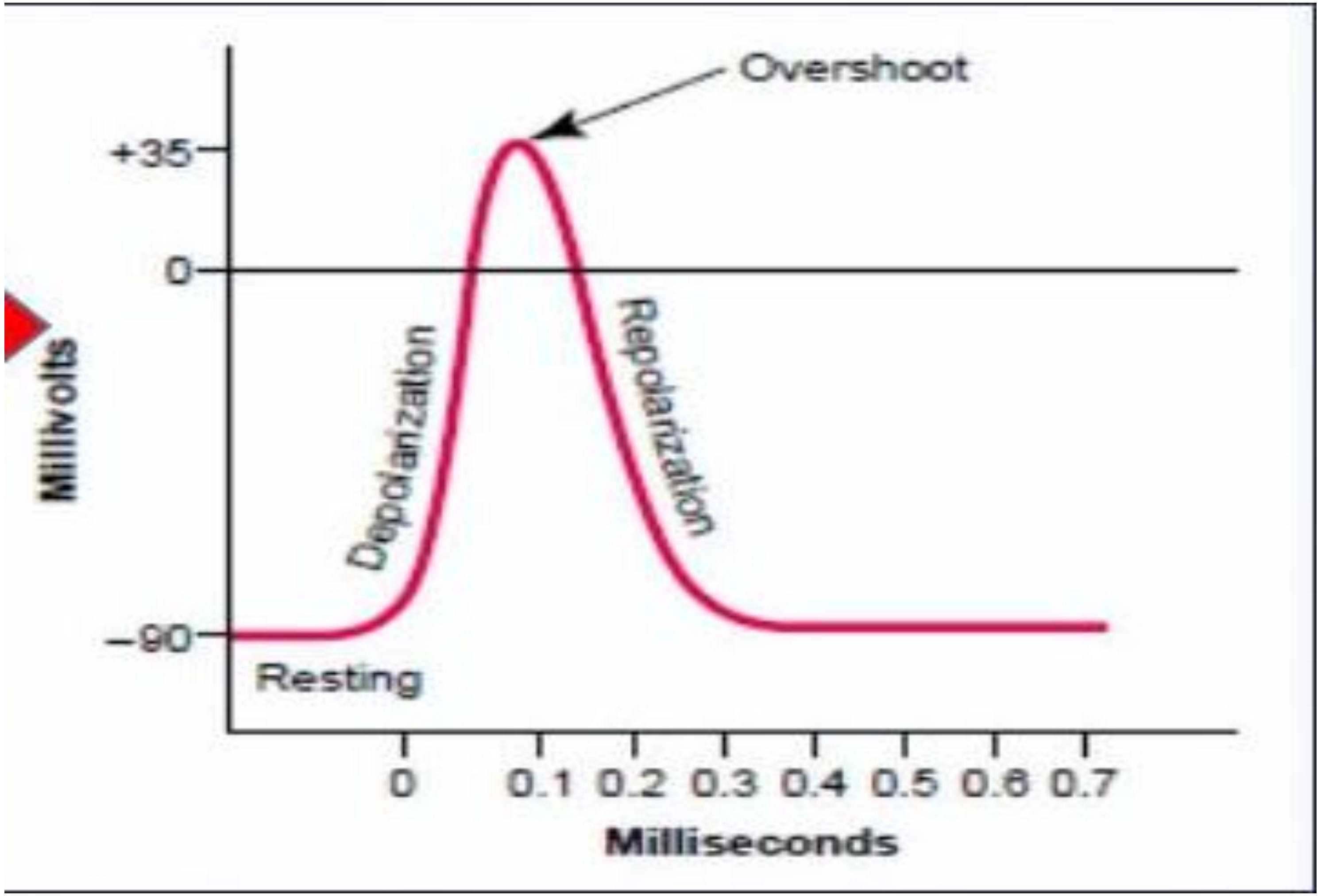
ACTION POTENTIAL

- When stimulation is given there is rapid changes in the membrane potential that spread rapidly along the nerve fiber membrane.
- These are Nerve signals
- Each action potential begins with a sudden change from the normal resting negative membrane potential to a positive potential
- Ends with an almost equally rapid change back to the negative potential.

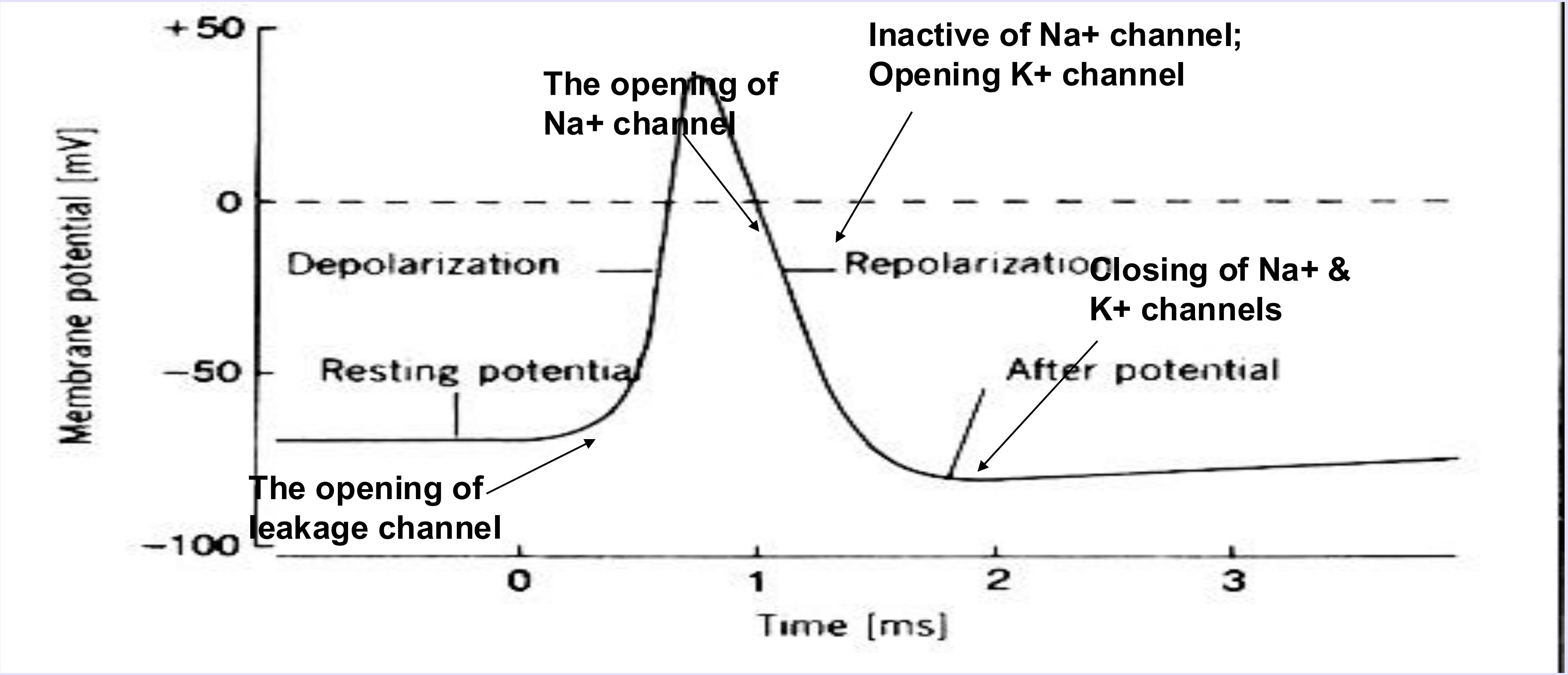


stages of the action potential

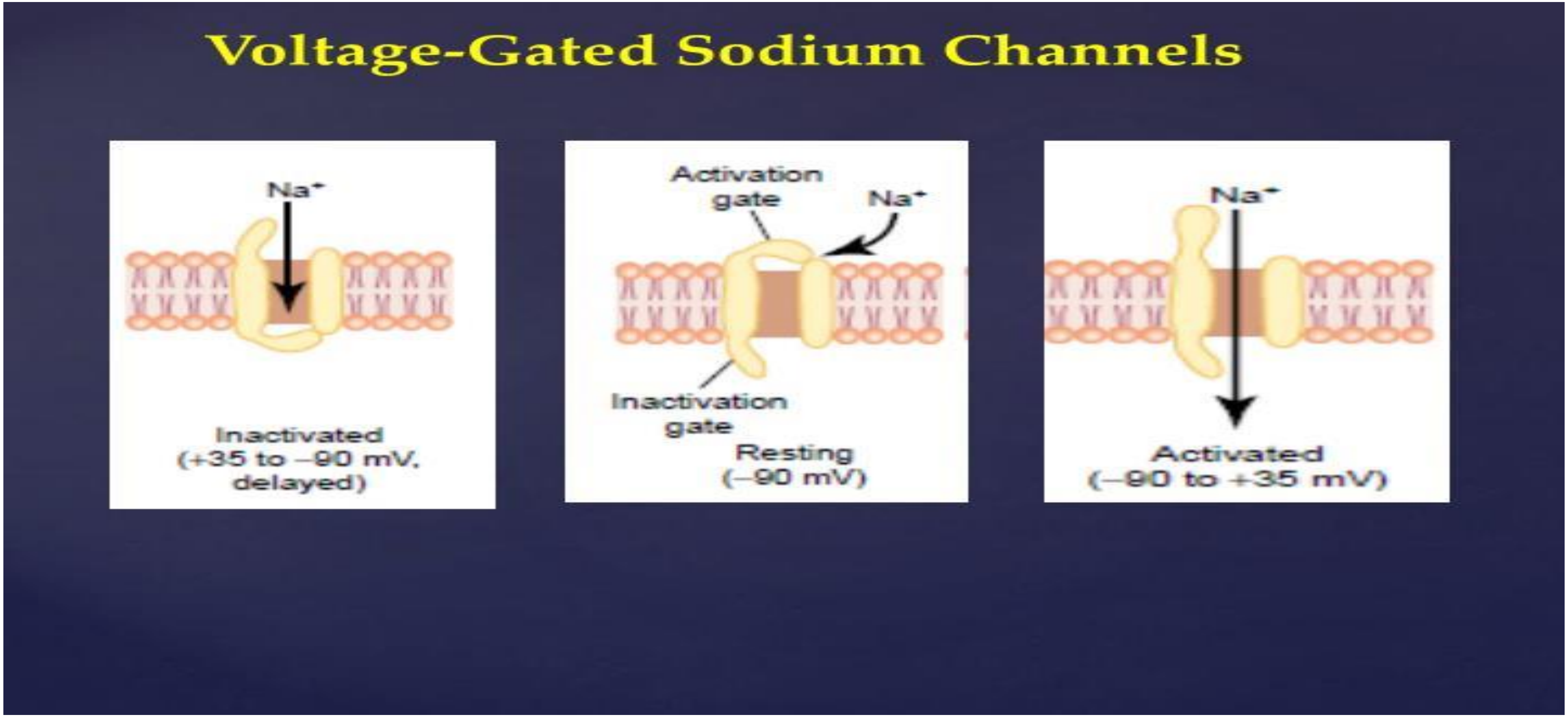
- Resting stage
- Depolariation stage
- Repolarization stage
- hyperpolarization



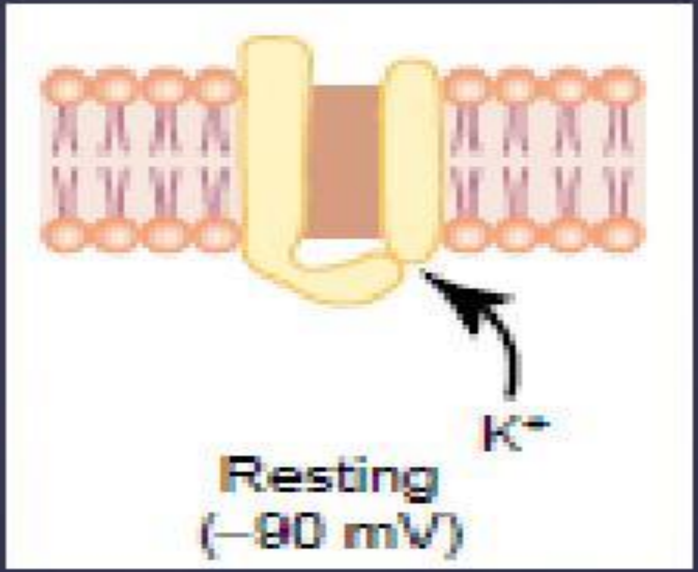
CHANNELS DURING ACTION POTENTIAL

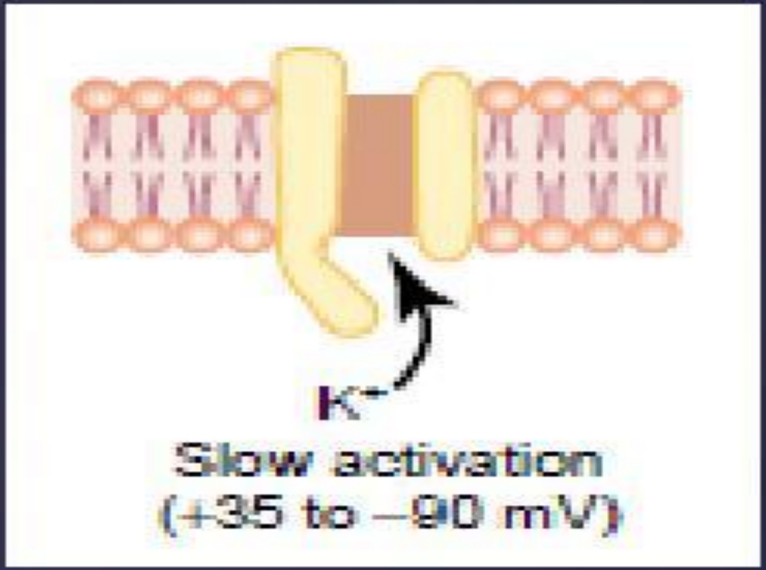


Activation and Inactivation of the Voltage-Gated Sodium Channel

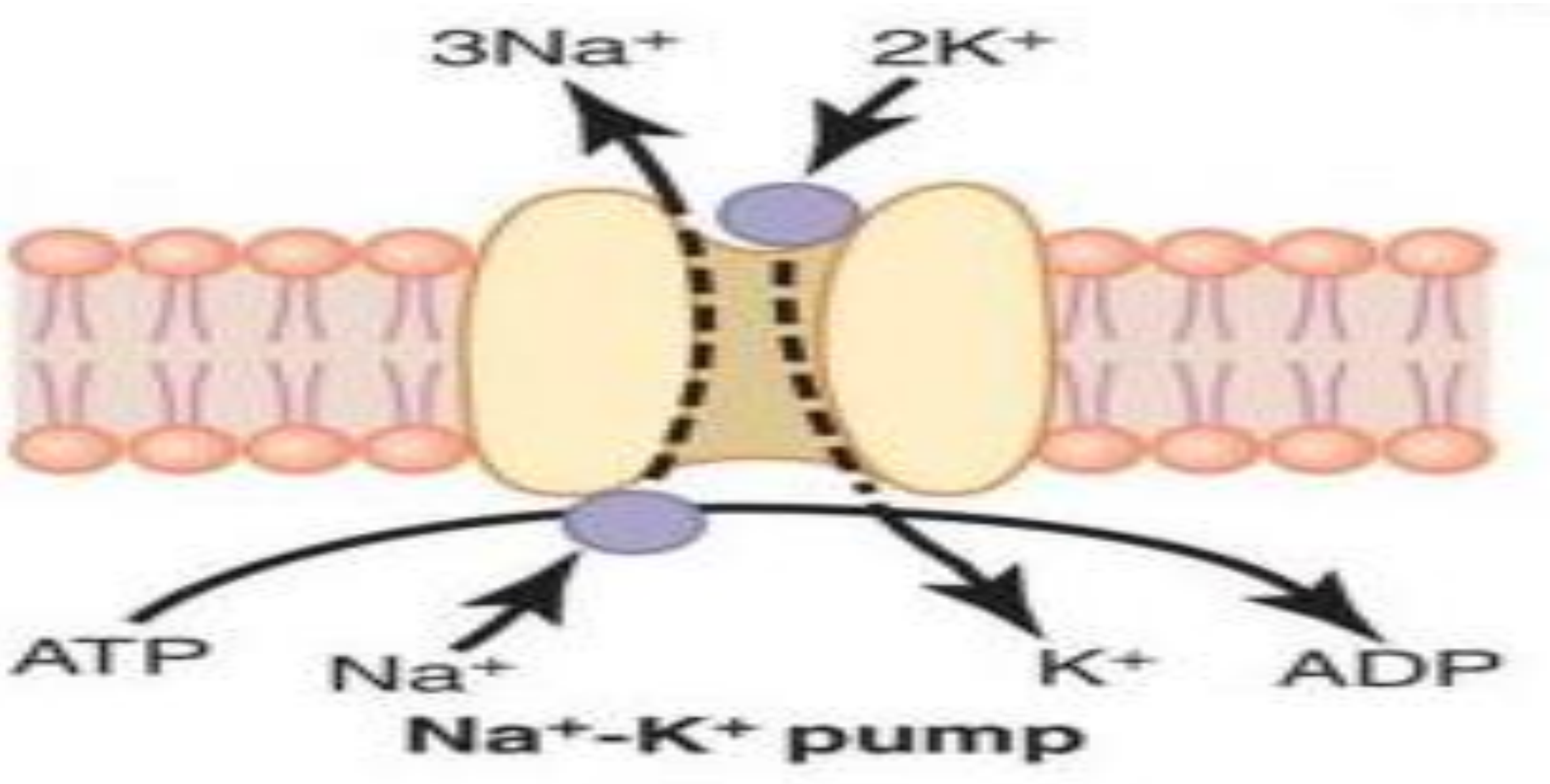


Voltage-Gated Potassium Channel





Re-establish RMP by Na-K pump



CHARACTERISTICS OF ACTION POTENTIALS

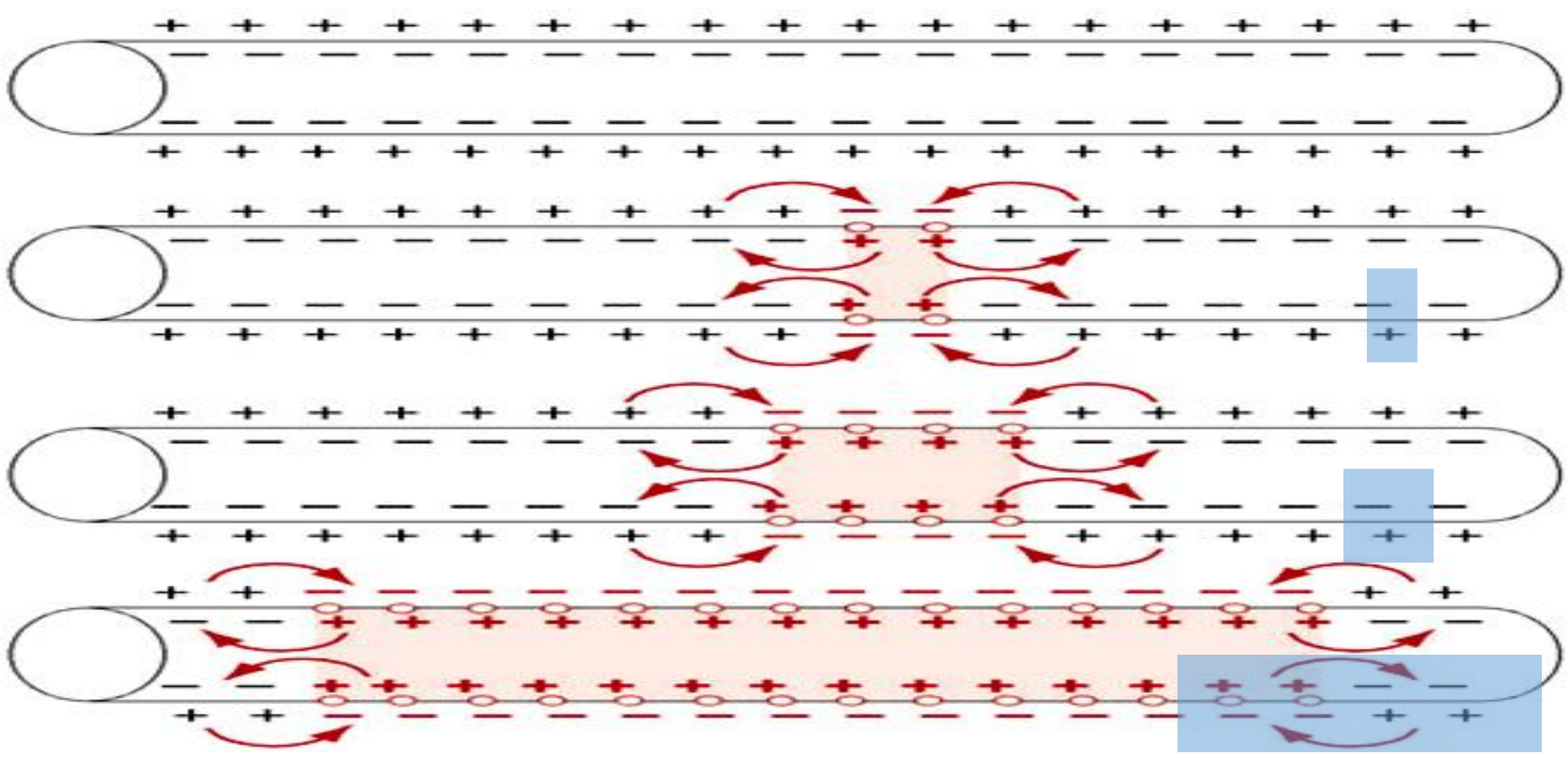
- Action potentials have three basic characteristics: stereotypical size and shape, propagation, and all-or-none response.
- **Stereotypical size and shape**. Each normal action potential for a given cell type looks identical, depolarizes to the same potential, and repolarizes back to the same resting potential.
- **Propagation**. An action potential at one site causes depolarization at adjacent sites, bringing those adjacent sites to threshold,(Self propagated). Propagation of action potentials from one site to the next is non decremental.
- **All-or-none response**. An action potential either occurs or does not occur. If an excitable cell is depolarized to threshold in a normal manner, then the occurrence of an action potential is inevitable. On the other hand, if the membrane is not depolarized to threshold, no action potential can occur.

Propagation of action potential

- Contiguous in unmyelinated nerve fibers
- Saltatory in myelinated nerve fibers

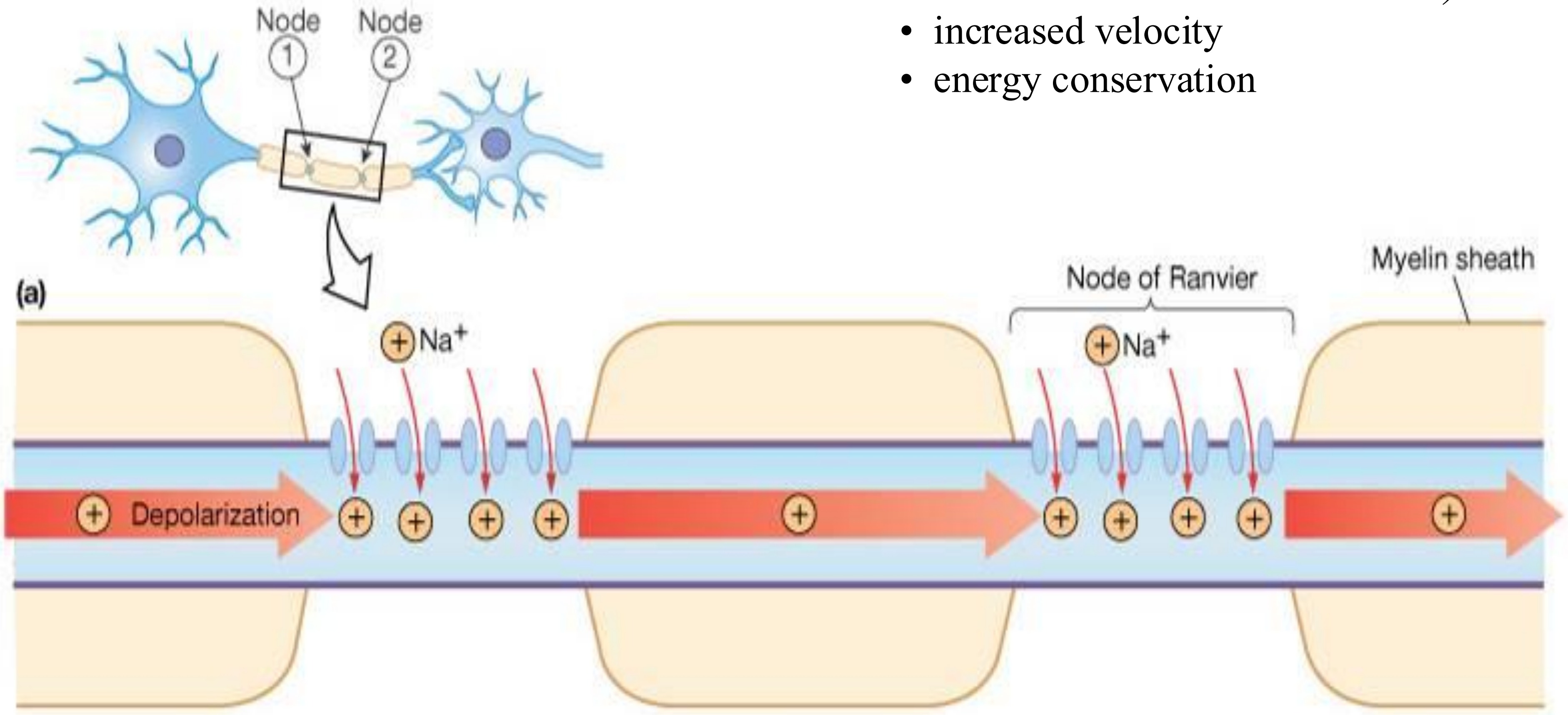
Contiguous Propagation:

Opening of Na^+ channels generates local current circuit that depolarizes adjacent membrane, opening more Na^+ channels…



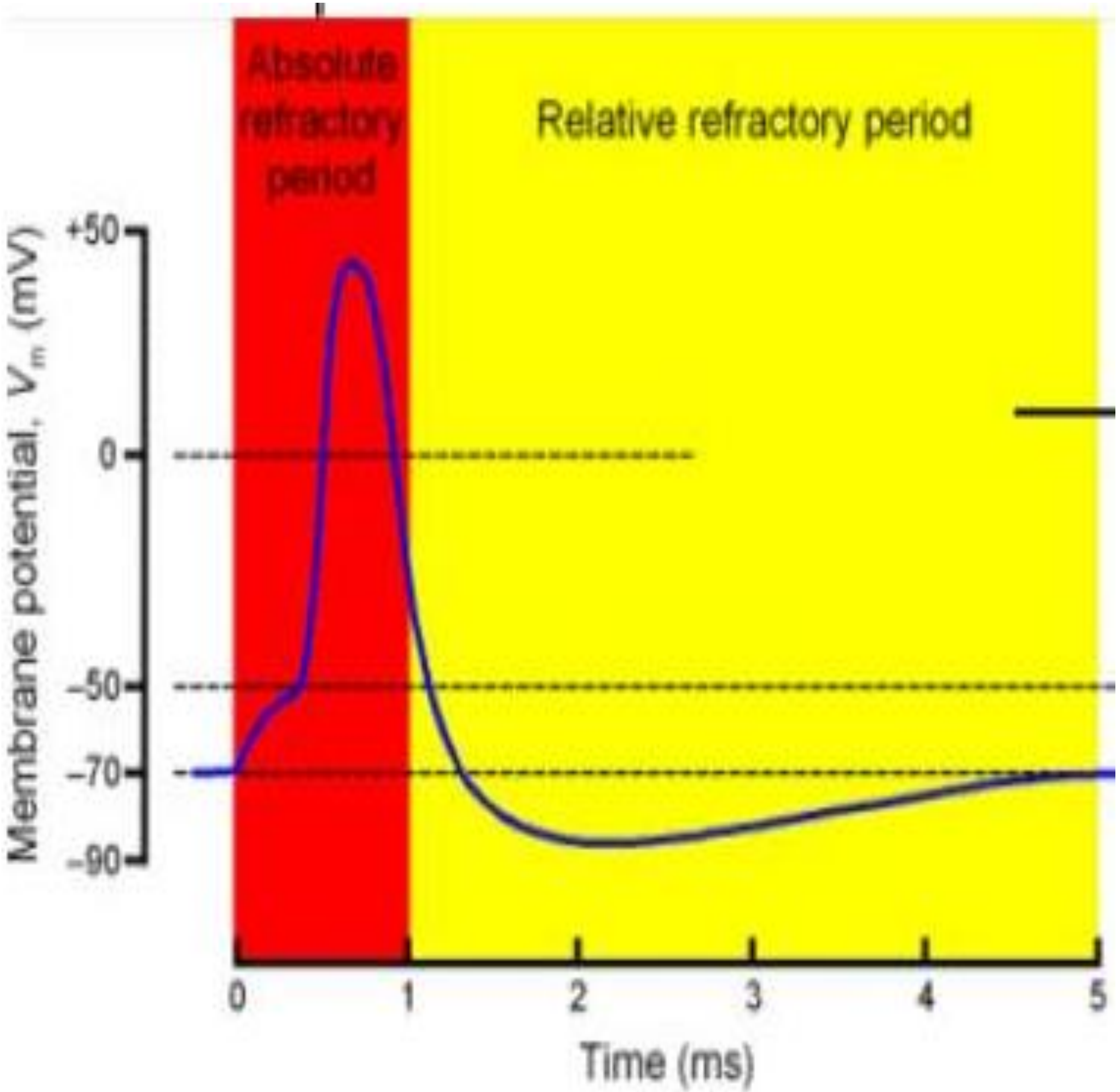
Saltatory conduction

- AP's only occur at the nodes *(Na channels concentrated here!)*
- increased velocity
- energy conservation

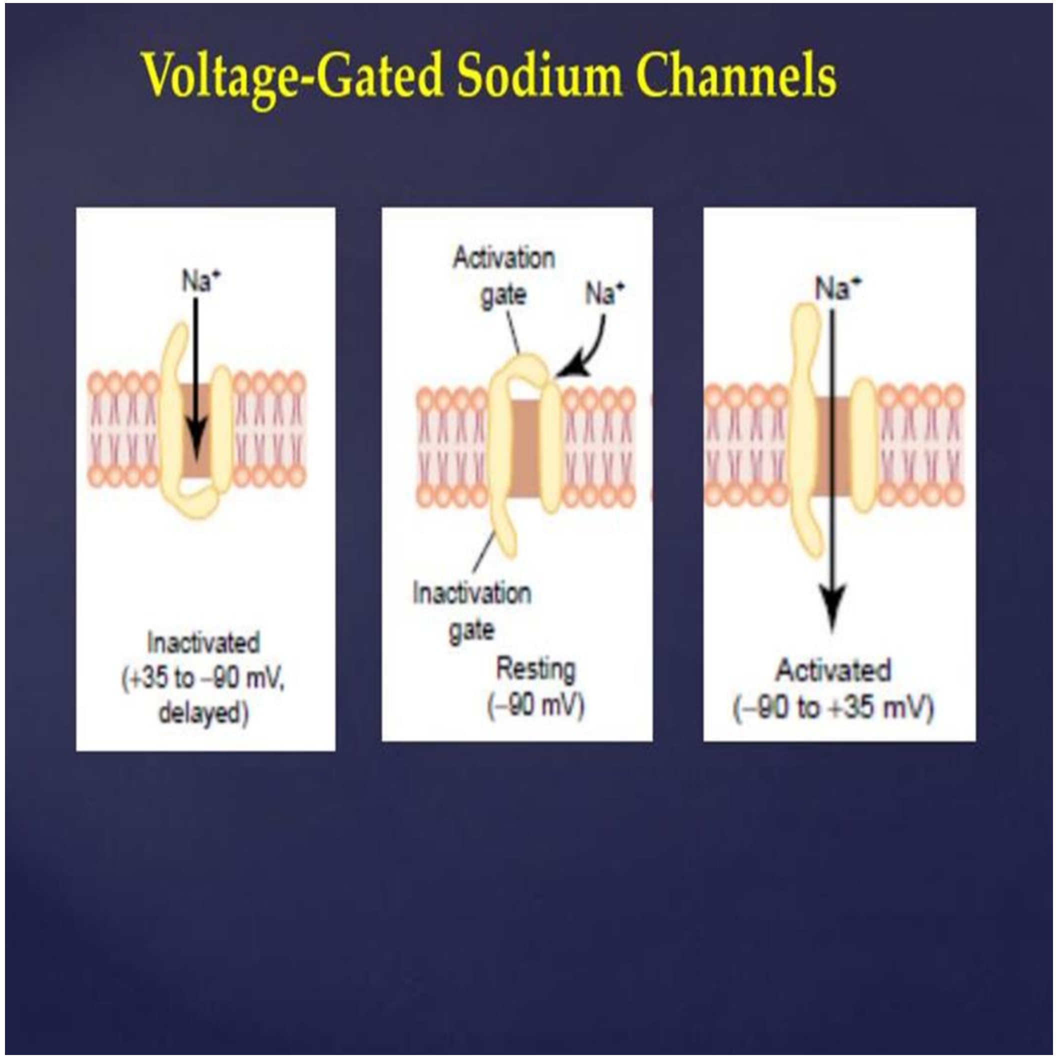


Refractory period

- A new action potential cannot occur in an excitable fiber as long as the membrane is still depolarized from the preceding action potential.
- The reason is that shortly after the action potential is initiated, the sodium channels become inactivated and no amount of excitatory signal applied to these channels at this point will open the inactivation gates
- The only condition that will allow them to reopen is for the membrane potential to return to or near the original resting membrane potential level



- Types of refractory period
- 2 types: Absolute & Relative
- Absolute: The period during which a second action potential cannot be elicited, even with a strong stimulus, is called the absolute refractory period.
- Relative: The period during which second action potential can be generated when membrane potential return to or near the original resting membrane potential level during which inactivation gate is opening




"If you cannot do great things.
do small things in a great way."
– NAPOLEON HILL